Physics
Standard and higher level
Paper 1A and Paper 2
B.3 Gas laws

12 October 2026

W4ss Academy practice test 2 | **Topic** B.3 Gas laws | **Ref.** W4SS–PHY–B3–02

Student name

30 minutes [Paper 1A and Paper 2]

Instructions to students

- Write your name in the boxes above.
- Do not open this examination paper until instructed to do so.
- Answer all questions.
- **Paper 1A:** for each question, choose the answer you consider to be the best and indicate your choice on the answer sheet provided.
- **Paper 2:** answers must be written within the answer boxes provided.
- A calculator is required for this paper.
- A clean copy of the **physics data booklet** is required for this paper.
- The maximum mark for paper 1A is **[10 marks]**.
- The maximum mark for paper 2 is **[10 marks]**.
- The maximum mark for this examination paper is **[20 marks]**.

Paper 1A

1. Which assumption is part of the molecular kinetic model of ideal gases?

A. The work done on a system equals the change in kinetic energy of the system.

B. The volume of a gas results from adding the volume of the individual molecules.

C. A gas is made up of tiny identical particles in constant random motion.

D. All particles in a gas have kinetic and potential energy.

2. An ideal gas is in a closed container. Which changes to its volume and temperature when taken together must cause a decrease in the gas pressure?

	Volume	Temperature
A.	decrease	increase
B.	decrease	decrease
C.	increase	increase
D.	increase	decrease

3. What is the relation between the value of the unified atomic mass unit in grams and the value of Avogadro's constant in mol^{-1}?

A. Their ratio is 1.

B. Their product is 1.

C. Their sum is 1.

D. Their difference is 0.

4. What is true for an ideal gas?

A. $nRT = Nk_BT$

B. $nRT = k_BT$

C. $RT = Nk_BT$

D. $RT = k_BT$

5. Gases in the atmosphere are compounds of $^{12}_{6}C$, $^{1}_{1}H$, $^{16}_{8}O$ and $^{14}_{7}N$.

Four of these gases are CO_2, N_2O, CH_4 and H_2O. A pure sample of each gas is produced. Each sample has the same mass.

Which sample contains the greatest number of molecules?

A. N_2O

B. H_2O

C. CO_2

D. CH_4

6. A sample of oxygen gas with a volume of 3.0 m^3 is at 100 °C. The gas is heated so that it expands at a constant pressure to a final volume of 6.0 m^3. What is the final temperature of the gas?

A. 750 °C

B. 470 °C

C. 370 °C

D. 200 °C

7. A fixed mass of an ideal gas expands slowly at constant temperature in a container.

Three statements about the gas molecules during the expansion are:

I. They collide with the walls of the container at a reduced rate.
II. They travel further on average between each collision.
III. Their average kinetic energy decreases as the gas expands.

Which statements are correct?

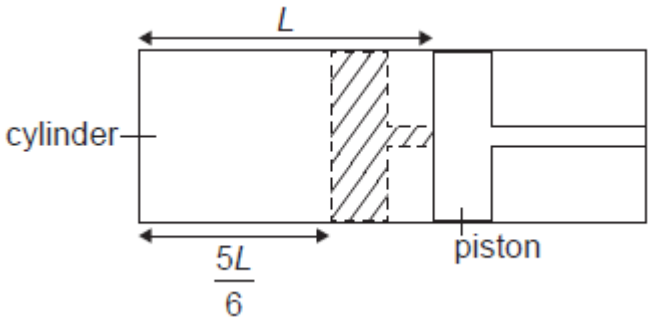
A. I and II only

B. I and III only

C. II and III only

D. I, II and III

8. A quantity of an ideal gas is at a temperature T in a cylinder with a movable piston that traps a length L of the gas. The piston is moved so that the length of the trapped gas is reduced to $\frac{5L}{6}$ and the pressure of the gas doubles.



What is the temperature of the gas at the end of the change?

A. $\frac{5}{12}T$

B. $\frac{3}{5}T$

C. $\frac{5}{3}T$

D. $\frac{12}{5}T$

9. Two containers X and Y are maintained at the same temperature. X has volume 4 m^3 and Y has volume 6 m^3. They both hold an ideal gas. The pressure in X is 100 Pa and the pressure in Y is 50 Pa. The containers are then joined by a tube of negligible volume. What is the final pressure in the containers?

A. 70 Pa

B. 75 Pa

C. 80 Pa

D. 150 Pa

10. A balloon of volume V contains 10 mg of an ideal gas at a pressure P. An additional mass of the gas is added without changing the temperature of the balloon. This change causes the volume to increase to $2V$ and the pressure to increase to $3P$.

What is the mass of gas **added** to the balloon?

A. 5 mg

B. 15 mg

C. 50 mg

D. 60 mg

Paper 1A — answer sheet

Student name _____ Date _____

1.	A	B	C	D
2.	A	B	C	D
3.	A	B	C	D
4.	A	B	C	D
5.	A	B	C	D
6.	A	B	C	D
7.	A	B	C	D
8.	A	B	C	D
9.	A	B	C	D
10.	A	B	C	D

Shade **one** circle per question. If you change your answer, cross out the wrong circle clearly.

Paper 2

Answer **all** questions. Answers must be written within the answer boxes provided.

1. A container of volume 3.2 × 10-6 m^3 is filled with helium gas at a pressure of 5.1 × 10^5 Pa and temperature 320 K. Assume that this sample of helium gas behaves as an ideal gas.

A helium atom has a volume of 4.9 × 10^{-31} m^3.

(a) The molar mass of helium is 4.0 g mol^{-1}. Show that the mass of a helium atom is 6.6 × 10^{-27} kg. [1]

(b) Estimate the average speed of the helium atoms in the container. [2]

(c) Show that the number of helium atoms in the container is about 4 × 10^{20}. [2]

(d) (i) Calculate the ratio $\frac{\text{total volume of helium atoms}}{\text{volume of helium gas}}$. [1]

(This question continues on the following page)

(Question 1 continued)

(ii) Explain, using your answer to (d)(i) and with reference to the kinetic model, why this sample of helium can be assumed to be an ideal gas. [2]

2. The average speed of the molecules of a gas is 500 m s^{-1}. The density of the gas is 1.2 kg m^{-3}. Calculate, in kPa, the pressure of the gas. [2]

Sources of the questions

This practice test was put together by W4ss Academy for internal class use. The questions are taken from the IB DP Physics Questionbank (first assessment 2025), topic B.3 Gas laws. They remain the property of the International Baccalaureate Organization and must not be published or sold.

Paper 1A — question 1	22M.1A.SL.TZ2.12
Paper 1A — question 2	19N.1A.SL.TZ0.10
Paper 1A — question 3	21M.1A.SL.TZ1.25
Paper 1A — question 4	22M.1A.SL.TZ2.11
Paper 1A — question 5	22N.1A.SL.TZ0.11
Paper 1A — question 6	21M.1A.SL.TZ2.10
Paper 1A — question 7	23M.1A.SL.TZ2.9
Paper 1A — question 8	22M.1A.SL.TZ2.10
Paper 1A — question 9	20N.1A.SL.TZ0.9
Paper 1A — question 10	23M.1A.SL.TZ2.10
Paper 2 — question 1	19M.2.SL.TZ2.2
Paper 2 — question 2	EXE.2.SL.TZ0.7 (part (c))

Please **do not** write on this page.

Answers written on this page
will not be marked.